

MINI PROJECT I

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WHAT IS THE QUESTION?

- Can word frequency reveal an author's style?
- Are books by the same author more similar than books from different authors?
- I compared *The Great Gatsby*, *This Side of Paradise* (F. Scott Fitzgerald) to *Frankenstein* (Mary Shelley).

WHAT WAS THE APPROACH?

- Used requests to pull all three texts from Project Gutenberg.
- Cleaned HTML markup, lowercased, removed punctuation, and removed stop words.
- Counted word frequencies per book, then normalized to the length of the book so books could be more fairly compared.
- Compared Paradise vs Gatsby (Fitzgerald) and Fitzgerald vs. Shelley by looking at the gaps in normalized frequency.

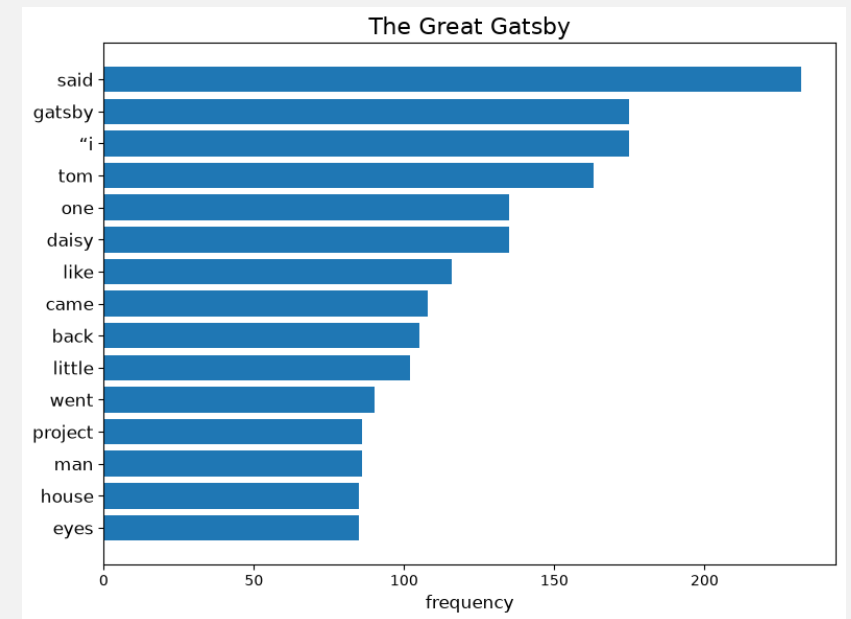
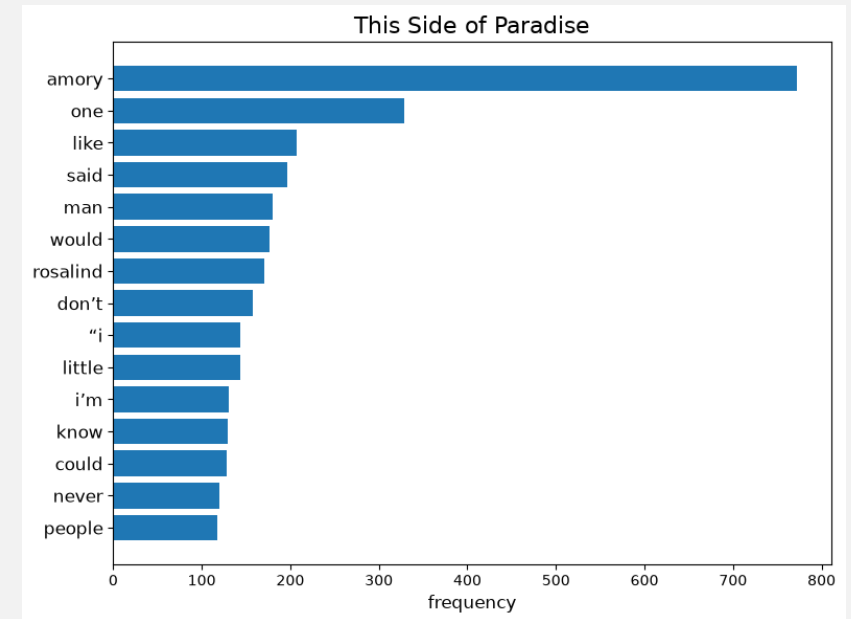
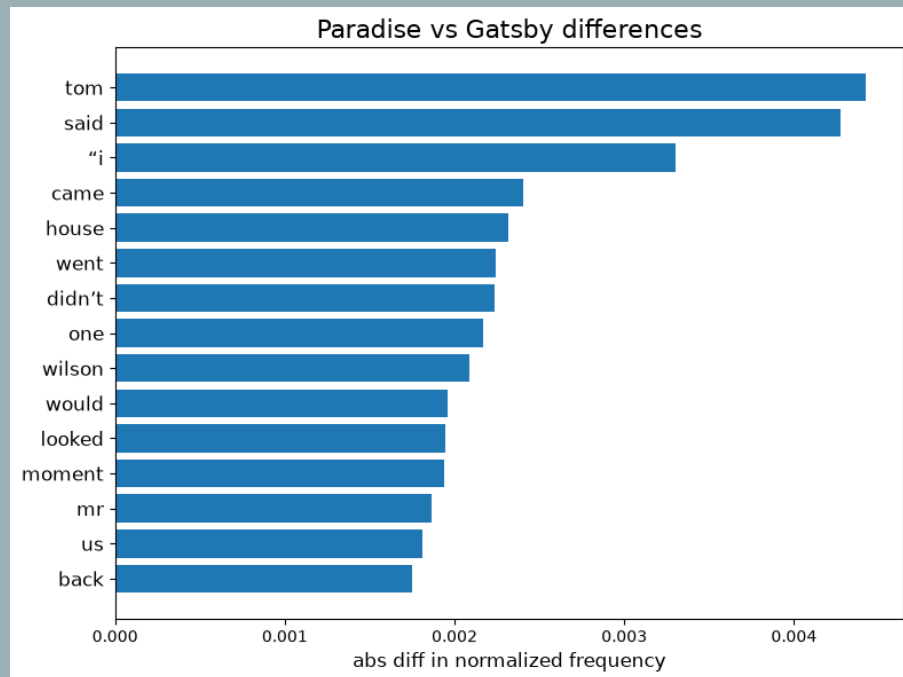
WHAT PROBLEMS DID I ENCOUNTER?

- NLTK's stopword list needed to be downloaded first before things would run.
- Raw frequency counts weren't comparable since the books all have different lengths making the data off.
- Gutenberg's format includes a lot of extra stuff (links, footers, etc) that cause noise if not removed.

WHAT RESULTS DID I GET?

Same author (Paradise vs. Gatsby):

Fairly similar although character names were notably different.

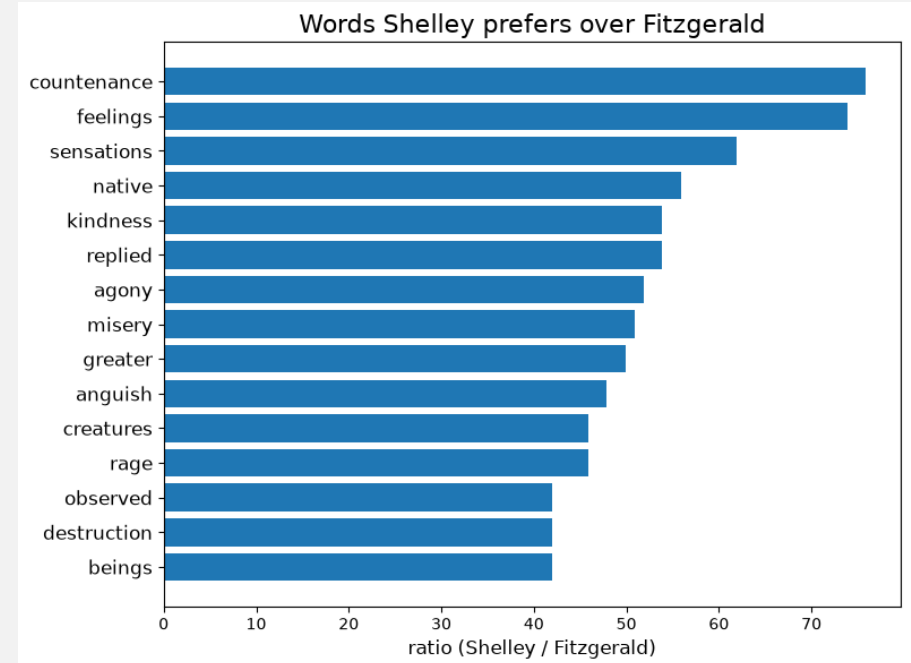
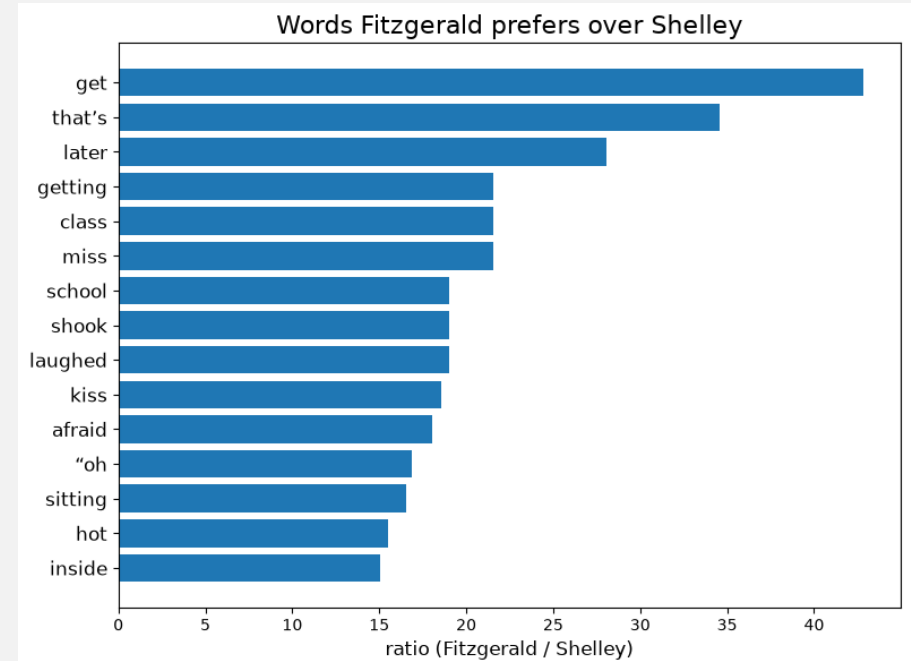
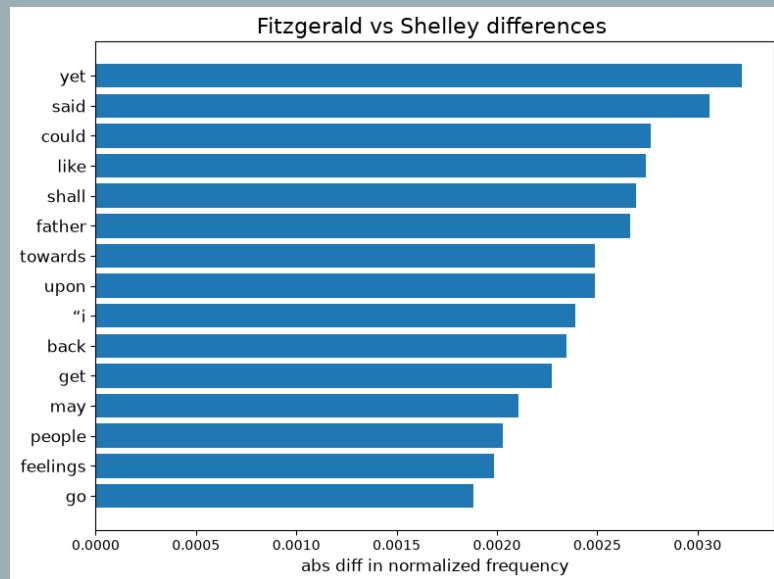


WHAT RESULTS DID I GET?

Different author (Fitzgerald vs. Shelley):

Shelley uses more dated language like “yet”, “upon” and tends to use longer words (as referenced by Words Shelley prefers over Fitzgerald)

Fitzgerald uses more modern language being a more recent author, “said” “like”.



WHAT NEW IDEAS DID THIS GENERATE?

- The frequency gap between the same author tends to be a lot smaller (but skewed a little by names or proper nouns in the writings). There's bigger gaps between different authors which shows that word-usage patterns can somewhat represent an author's style.
- Word choices are very different and follow when books are written. Shelley lived in the early 1800s and wrote gothic literature while Fitzgerald wrote mainly during and about the Jazz age. However, I wonder if this still applies if authors try to write outside of their time/style?